CHAIN OF CUSTODY FORM
(Original Stays With Samples)

US EPA Experimental Stream Facility
1003 US Highway 50, Milford, Ohio 45150
Phone # 513-965-4805
FAX # 513-965-4806

Originator:	
Ship Date: 20180215	Contact:
Destination: Cincinnati	Phone #:
Recipient: Sam Smith	E-Mail:
Phone #: 513-569-7681	

Collection Date YYYY-MM-DD	Collection Time HH:MM:SS	SiteID	LongID	CrossID	REP	Matrix Type*	TYPE (UKN, DUP, SPK, BLK)	Preservative	Analyses Required
20180208	16:05:00				1	DI	BTL BLK		TNO3, TNO2, TRP, INH4, TNP, TPTN
20180208	16:10:00				1	DI	BTL BLK		"
20180208	16:15:00				1	DI	BTL BLK		"
20180212	12:29:00				1	DI	FIELD BLK		"
20180213	09:35:00	FL	02	0.1	1	SW	UNK		"
20180213	09:50:00	FL	05	0.1	1	SW	UNK		"
20180213	10:15:00	FL	01	0.1	1	SW	UNK		"
20180213	10:35:00	FL	07	0.1	1	SW	UNK		"
20180213	11:10:00	FL	15	0.1	1	SW	UNK		"
20180213	11:30:00	FL	14	0.1	1	SW	UNK		"
20180213	11:40:00	FL	03	0.1	1	SW	UNK		"
20180213	11:55:00	FL	04	0.1	1	SW	UNK		"
20180213	12:15:00	FL	45	0.1	1	SW	UNK		"
20180213	12:40:00	FL	49	0.1	1	SW	UNK		"
20180213	13:10:00	FL	43	0.1	1	SW	UNK		"
20180213	13:35:00	FL	47	0.1	1	SW	UNK		"
20180214	14:30:00				1	DI	HOUSE BLK		"
20180208	16:05:00				1	DI	BTL BLK	2M HCl	TOC
20180208	16:10:00				1	DI	BTL BLK	"	"
20180208	16:15:00				1	DI	BTL BLK	"	"
20180212	12:29:00				1	DI	FIELD BLK	"	"
20180213	09:35:00	FL	02	0.1	1	SW	UNK	"	"
20180213	09:50:00	FL	05	0.1	1	SW	UNK	"	"
20180213	10:15:00	FL	01	0.1	1	SW	UNK	"	"
20180213	10:35:00	FL	07	0.1	1	SW	UNK	"	"
20180213	11:10:00	FL	15	0.1	1	SW	UNK	"	"
20180213	11:30:00	FL	14	0.1	1	SW	UNK	"	"
20180213	11:40:00	FL	03	0.1	1	SW	UNK	"	"
20180213	11:55:00	FL	04	0.1	1	SW	UNK	"	"
20180213	12:15:00	FL	45	0.1	1	SW	UNK	"	"
20180213	12:40:00	FL	49	0.1	1	SW	UNK	"	"
	Relinquished By:		Date:	Time:	Received By:		Date:		Time:

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Collection Date YYYY-MM-DD	Collection Time HH:MM:SS	SiteID	LongID	CrossID	REP	Matrix Type*	TYPE (UKN, DUP, SPK, BLK)	Preservative	Analyses Required
20180213	13:10:00	FL	43	0.1	1	SW	UNK	2M+HCl	TOC
10180213	13:35:00	FL	47	0.1	1	SW	UNK	2M+HCl	TOC
20180214	14:30:00				1	DI	HOUSE BLK	2M+HCl	TOC

Relinquished By:	Date:	Time:	Received By:	Date:		Time:

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(Original Stays With Samples)

US EPA Experimental Stream Facility
1003 US Highway 50, Milford, Ohio 45150
Phone # 513-965-4805
FAX # 513-965-4806

Originator:	
Ship Date:	
Destination:	
Recipient:	
Phone #:	

Contact:
Phone #:
E-Mail:

Collection Date YYYY-MM-DD	Collection Time HH:MM:SS	SiteID	LongID	Cross ID	REP	Matrix Type*	TYPE (UKN, DUP, SPK, BLK)	Analyses Required	*Volume of liquid sample filtered (mL)	
20180213	09:35:00	FL	02	0.1	1	SW	UNK	Chlorophyll	335.49	
20180213	09:50:00	FL	05	0.1	1	SW	UNK	"	315.58	
20180213	10:15:00	FL	01	0.1	1	SW	UNK	"	287.26	
20180213	10:35:00	FL	07	0.1	1	SW	UNK	"	285.95	
20180213	11:10:00	FL	15	01	1	SW	UNK	"	295.18	
20180213	11:30:00	FL	14	0.1	1	SW	UNK	"	217.99	
20180213	11:40:00	FL	03	0.1	1	SW	UNK	"	271.92	
20180213	11:55:00	FL	04	0.1	1	SW	UNK	"	192.52	
20180213	12:15:00	FL	45	0.1	1	SW	UNK	"	200.48	
20180213	12:40:00	FL	49	0.1	1	SW	UNK	"	160.84	
20180213	13:10:00	FL	43	0.1	1	SW	UNK	"	102.16	
20180213	13:35:00	FL	47	0.1	1	SW	UNK	"	121.92	
20180212	12:29:00				1	BL	FIELDBLK	"	389.97	

Relinquished By:	Date:	Time:	Received By:	

GAS TRAP DATA SHEET

Lake:	FALLS		2/13/18	2/14/18		
Site	**Lat**	**Long**	**Deployment time/date**	**Retrieval time/date**	**Volume recovered (mL)**	**Exetainer codes**
24	35.941929	78.588048	9:20	9:00	0	
16	35.956840	78.590437	9:25	9:15	5	
2	35.976222	78.588994	9:35	9:30	0	
5	35.970107	78.621309	9:50	9:45	3	
12	35.965770	78.634064	9:55	9:55	0	
13	36.003475	78.646342	10:10	10:10	1	
1	36.002145	78.634479	10:15	10:20	0	
21	36.008775	78.628127	10:20	10:28	0	
7	36.005622	78.667541	10:35	10:45	0	VAlve Open
11	36.019263	78.717061	11:00	11:00	1	
15	36.022915	78.714751	11:10	11:10	7	
14	36.059410	78.717674	11:30	11:20	0	
3	36.032185	78.736782	11:40	11:40 ~~11:48~~	~~8~~ 0	
8	36.018850	78.748857	11:50	11:50	1	Trap Just After Sample collection.
4	36.012870	78.754494	11:55	11:55	0	
10	36.039907	78.756001	12:05	12:10	0	
45	36.044152	78.758162	12:15	12:20	0	
57	36.048510	78.752361	12:30	12:30	0	
49	36.051345	78.757112	12:40	12:40	2	
61	36.053488	78.759497	12:45	12:50	1	
55	36.053250	78.761884	12:50	12:57	9	
54	36.057853	78.765296	12:55	13:05	0	
63	36.055987	78.770364	13:05	13:15	0	
43	36.061542	78.773187	13:10	13:25	0	
59	36.062867	78.771126	13:20	13:35	9	
58	36.068647	78.775921	13:25	13:45	0	
47	36.068365	78.768794	13:35	13:55	7	
08	36.069257	78.766001	13:40	14:05	4	
56	36.072975	78.766212	13:45	14:15	0	
52	36.079438	78.760791	13:55	14:25	2	

DISSOLVED GAS AND AIR DATA SHEET

Lake:	Falls				Date: 2/14/18			
¥Site	Dissolved gas Exetainer codes 02142018.–		Water Vol/ Helium Vol	Water Temp (degC)	Atmospheric Air Exetainer Codes			BP (mm Hg)
2	Fl.02.DG.1	Fl.02.DG.2	115/125	8.9	02142018.Fl	.02.AA.1,	Fl02.AA.2,	Fl.02.AA.3.
5	Fl.05.DG.1	Fl.05.DG.2		10.0				
1	Fl.01.DG.1	Fl.05.DG2		11.2				
7	Fl.07.DG.1	Fl.07.DG.2		12.1				
15	Fl.15.DG.1	Fl.15.DG.2		11.5				
14	Fl.14.DG.1	Fl.14.DG.2		10.9				
3	Fl.03.DG.1	Fl.03.DG.2		11.5				
4	Fl.04.DG.1	Fl.04.DG.2		12.2				
45	Fl.45.DG.1	Fl.45.DG.2		13.5				
49	Fl.49.DG.1	Fl.49.DG.2		13.5	Vial 2	may have	small gas	volume
43	Fl.43.DG.1	Fl.43.DG.2		12.8	Fl.43.AA.1	Fl.43.AA.2	Fl.43.AA.3	
47	Fl.47.DG.1	Fl.47.DG.2		12.3				

LGR Ambient @ 13:30:00

¥Site: all dissolved gas samples collected at 0.1m depth

Boat motor should be off for 2 minutes before sample collection

Water Vol/Air Vol :: 115ml/25ml

Goal is to inject 20mL of gas into exetainer

Signature Date

GREENHOUSE GAS ANALYZER AND GPS DATA SHEET

FILE NAMES	
Lake:	FANS
Sample Date:	2/14/18
GGA file	GGA_2018-02-14_f0000.txt
gps file name	

Time GPS 10:21:00 LGR 10:19:30

DEPLOYMENT DATA

Sample site	Chamber start time	Bubbling observed (Y/N)	GPS logger started	Chamber ~~volume (L)~~ MARK	Water Temperature (C)	CH_4 and CO_2 profile checked
24	8:59:30	✓	✓	55	8.2	✓
16	9:14:00	✓	✓	55	8.6	✓
2	9:29:45	✓	✓	45	8.9	✓
5	9:43:25	✓	✓	45	9.3	✓
12	9:53:24	✓	✓	45	9.1	✓
13	10:08:16	✓	✓	45	9.6	✓
1	10:17:52	✓	✓	45	10.1	✓
71	10:26:20	✓	✓	50	10.0	✓
7	10:43:50	✓	✓	50	9.4	✓
11	10:58:47	✓	✓	50	10.3	✓
15	11:07:11	✓	✓	50	9.9	✓
14	11:20:10	✓	✓	50	10.9	✓
3	11:37:00	✓	✓	50	10.3	✓
8	11:48:30	✓	✓	50	10.0	✓
41	11:57:11	✓	✓	50	10.1	✓
10	12:10:10	✓	✓	50	11.2	✓
45	12:20:10	✓	✓	50	11.5	✓
57	12:31:13	✓	✓	50	11.5	✓
49	12:41:00	✓	✓	50	11.2	✓
61	12:49:17	✓	✓	50	11.4	✓
55	12:56.00	✓	✓	50	11.4	✓
54	13:05:35	✓	✓	50	11.3	✓
63	13:14:42	✓	✓	50	11.1	✓
43	13:25:50	✓	✓	50	10.9	✓
59	13:34:15	✓	✓	50	10.8	✓
58	13:45:15	✓	✓	50	10.5	✓
47	13:53:54	Yes	✓	50	10.8	✓
47	13:58:43	✓	✓	50	10.8	✓
68	14:06:30	✓	✓	50	10.9	✓
56	14:15:40	✓	✓	50	10.9	✓
52	14:24:55	N	✓	50	11.0	✓

SONDE DATA SHEET

Lake: Sample Date:

Site	Site Depth (m)	Sample Depth (m)	Temp (C)	DO (mg/l)	DO (%sat)	sp.Cond (µs cm^{-1})	pH	Chl a (ug/L)	Turbidity (NTU)	Notes
24	3.0	0.1	8.55	11.2	95.7	0.088	7.72	7.8	4.0	
14	5.3	0.1	9.09	12.2	105.3	0.086	7.71	10.8	7.7	
2	3.2	0.1	8.54	12.0	101.7	0.088	7.65	8.9	5.6	
5	4.2	0.1	8.93	12.7	109.8	0.092	7.64	14.0	4.1	
12	4.8	0.1	8.91	12.35	106.5	0.099	7.71	11.8	5.7	
13	4.2	0.1	8.97	12.73	109.8	0.097	7.77	15.6	6.1	
1	4.7	0.1	9.28	12.56	109.4	0.092	7.74	9.2	8.6	
21	4.2	0.1	9.81	12.17	106.5	0.084	7.69	13.7	18.8	
7	7.5	0.1	9.30	12.45	108.5	0.101	7.66	8.9	6.1	
↓	↓	1.0	9.38	12.42	108.4	0.104	7.68	22.6	4.9	
		2.0	9.38	12.15	106.1	0.104	7.69	25.0	4.8	
		3.0	9.32	12.02	104.6	0.104	7.[illegible]	25.0	4.8	
		4.0	9.29	11.85	103.3	0.104	7.70	26.1	4.8	
		5.0	9.21	11.79	102.5	0.104	7.70	26.5	5.0	
		~~6.0~~	~~8.60~~	~~11.53~~	~~9~~	Cl				
		6.0	8.49	10.97	93.5	0.103	7.70	23.5	5.3	
↓	↓	7.0	7.90	10.50	88.0	0.104	7.68	20.4	7.8	
11	3.3	~~9.5~~ 0.1	~~12.851~~	~~12.00~~	103.3	0.136	~~7.52~~	~~13.0~~ 10.5	6.9	
15	3.9	~~0~~ 0.1	9.48	11.92	104.0	0.136	7.59	14.0	7.3	
14	1.2	0.1	10.45	12.2	108.0	0.109	7.60	12.4	16.1	
3	1.4	0.1	10.03	12.16	107.5	0.192	7.51	12.3	12.1	
8	3.5	0.1	9.67	12.0	105.2	0.192	7.60	10.5	12.6	
4	1.3	0.1	9.55	12.35	108.5	0.186	7.60	9.9	15.2	
10	1.3	0.1	10.25	11.76	104.5	0.203	7.55	13.3	20.6	
45	2.1	0.1	10.40	11.96	105.8	0.205	7.59	12.1	21.4	

SONDE DATA SHEET

Lake: Sample Date:

Site	Site Depth (m)	Sample Depth (m)	Temp (C)	DO (mg/l)	DO (%sat)	sp.Cond (µs cm^{-1})	pH	Chl a (ug/L)	Turbidity (NTU)	Notes
57	1.8	0.1	10.4	11.87	108.1	0.198	7.54	15.9	27.7	
49	2.4	0.1	10.44	11.9	106.0	0.190	7.63	11.0	28.0	
61	2.1	0.1	10.40	11.8	104.7	0.188	7.66	13.5	35.9	
55	2.2	0.1	10.44	11.56	103.0	0.189	7.63	11.6	36.4	
54	1.6	0.1	10.72	10.75	96.8	0.172	7.61	12.0	56.6	
63	2.0	0.1	10.64	11.39	101.2	0.172	7.59	11.8	62.6	
48	1.9	0.1	10.79	10.30	93.0	0.169	7.57	11.4	70.7	
59	2.3	0.1	10.63	11.14	99.5	0.175	7.58	11.8	58.3	
58	1.5	0.1	10.48	11.16	100.0	0.163	7.55	11.3	76.7	
47	1.9	0.1	10.7	10.67	96.1	0.176	7.6	12.0	59.5	
68	2.0	0.1	10.69	11.18	100.7	0.173	7.55	13.5	55.3	
56	2.0	0.1	10.68	10.75	97.1	0.180	7.51	18.7	45.2	
52	1.2	0.1	10.87	10.55	95.5	0.184	7.47	12.3	42.0	

Date 2/12/18
Cal cylinder serial # CC163816
Dilution system serial # EPA A20808
LGR file 994 - 2018-02-12_f0002.txt

Zero Air Flow (Lpm)	Cal Gas Flow Readout (sccm)	CH4 cal (ppm)	CO2 cal (ppm)	CH4 resp (ppm)	CO2 resp (ppm)	Start Time	Stop Time
3.25	0.0	0.0	0.0	0.0136	0.07	8:22:20	8:24:46
9	7.0	2.4	23.7	2.420	24.21	8:26:00	8:31:00
8.5	7.0	2.5	25.1	2.570	25.56	8:31:00	8:34:30
8	7.0	2.7	26.7	2.72	27.04	8:34:30	8:36:54
7.5	7.0	2.8	28.4	2.88	28.80	8:36:54	8:39:54
7	7.0	3.0	30.5	3.08	31.00	8:39:54	8:42:08
6.5	7.0	3.3	32.8	3.31	33.25	8:42:08	8:44:38
6	7.0	3.6	35.5	3.59	35.90	8:44:38	8:47:51
5.5	7.0	3.9	38.8	3.92	34.24	8:47:51	8:51:50
5	7.0	4.3	42.7	4.31	43.06	8:51:50	8:53:40

Signature _____

Date 2/16/18
Cal cylinder serial # CC163816
Dilution system serial # EPA A20808
LGR file 999-2018-02-16_f0000.txt

Zero Air Flow (Lpm)	Cal Gas Flow Readout (sccm)	CH4 cal (ppm)	CO2 cal (ppm)	CH4 resp (ppm)	CO2 resp (ppm)	Start Time	Stop Time
3.25	0.0	0.0	0.0	0.012	-0.01	10:17:40	10:21:03
9	7.0	2.4	23.7	2.40	23.87	10:21:03	10:28:30
8.5	7.0	2.5	25.1	2.54	25.2	10:28:30	10:30:20
8	7.0	2.7	26.7	2.70	26.7	10:30:20	10:32:15
7.5	7.0	2.8	28.4	2.86	28.4	10:32:15	10:34:50
7	7.0	3.0	30.5				
6.5	7.0	3.3	32.8				
6	7.0	3.6	35.5				
5.5	7.0	3.9	38.8				
5	7.0	4.3	42.7				

Signature _____

ATTACHMENT A
YSI 6600 SONDE CALIBRATION LOG SHEET

SONDE ID: STD16 **DATE:** 2/13/18 **TECHNICIAN:** JTW Post 2/16/18

DATE / TIME OK? YES / NO

BATTERY VOLTS: 12.4 (Change if < 10.0 v) **CHANGED BATTERIES?** (YES) / NO

TURBIDITY WIPER: PARKS 180° FROM OPTICS? (YES) / NO **CHANGED?** YES / (NO)

CHLOROPHYLL WIPER: PARKS 180° FROM OPTICS? (YES) / NO **CHANGED?** YES / (NO)

DISSOLVED OXYGEN: "DO Sat %" **BAROMETRIC PRESSURE:** 748.4 mm Hg Post = 746.1

SENSOR OUTPUT TEST: PASS / FAIL (For Rapid-Pulse Probes, change membrane if failed)

Record first 10 readings from Discrete Sampling:

DO MEMBRANE CHANGED? YES / (NO) **PROBE RE-SURFACED?** YES / (NO) **DATE/TIME:** _____

[~97] **INITIAL:** 101.9 **CALIBRATED:** 100.7 **POST-DEPLOYMENT:** 98.0
DO CHARGE [50 ± 25]: _____ 7 **DO GAIN** [1.0 ± 0.35]: 1.04444

DEPTH: "Pressure" ~~[m]~~ ft **PRESSURE OFFSET** [VENTED = 0 ± 6 / NON-VENTED = -14.7 ± 6]: _____
[0] **INITIAL:** 0.478 **CALIBRATED:** 0.00 **POST-DEPLOYMENT:** -2.04

CONDUCTIVITY: "SpCond" [mS/cm] **CONDUCTIVITY CELL CONSTANT** [5.0 ± 0.45]: 4.89202 Pre
~~[10]~~ 1.413 @ 25 **INITIAL:** 1.404 **CALIBRATED:** 1.413 **POST-DEPLOYMENT:** ~~1.394~~ 1.413
Temp Pre = 22.38
Temp Post = 22.61

TURBIDITY: "Turb+NTU" [NTU]

[0] **INITIAL:** 0 **CALIBRATED:** 0 **POST-DEPLOYMENT:** -0.2
[1000] **INITIAL:** 125.1 **CALIBRATED:** 126 **POST-DEPLOYMENT:** 124.4

CHLOROPHYLL: "Chlor ug/L" [ug/L]

[0] **INITIAL:** -0.6 **CALIBRATED:** 0.2 **POST-DEPLOYMENT:** 0.0
Jan 23 S.Tek 43 ~~[113]~~ **INITIAL:** 2.9 **CALIBRATED:** → **POST-DEPLOYMENT:** 11.6 S.tk 43

---

pH: "pH"

[7] **INITIAL:** 6.99 **CALIBRATED:** 7.0 **POST-DEPLOYMENT:** 6.90 **MV** [0 ± 40]: -38.8 Pre Initial
[10] **INITIAL:** 10.05 **CALIBRATED:** 10.0 **POST-DEPLOYMENT:** 9.95 **MV** [-177 ± 40]: -214.2

ORP: "ORP mV" **ORP MV OFFSET [0 ± 100]:** _____

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